



# **Applied Data Mining**

**— Week 2—  
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# Outline

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- Data Objects and Attribute Types 
- Basic Statistical Descriptions of Data

# Data Objects

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- Data sets are made up of data objects.
- A **data object** represents an entity.
- Examples:
  - sales database: customers, store items, sales
  - medical database: patients, treatments
  - university database: students, professors, courses
- Also called *samples* , *examples*, *instances*, *data points*, *objects*, *tuples*.
- Data objects are described by **attributes**.
- Database (rows -> data objects; columns -> attributes)

# Attributes



- **Attribute** (or **dimensions**, **features**, **variables**): a data field, representing a characteristic or feature of a data object.
  - E.g., customer\_ID, name, address
- A set of attributes used to describe a given object is called an attribute vector (or feature vector).
- The distribution of data involving one attribute (or variable) is called uni-variate.
- A bi-variate distribution involves two attributes.

# Attributes



- Attribute Types:
  - Nominal (Related to Names)
  - Binary (Only two states)
  - Numeric: quantitative
    - Interval-scaled
    - Ratio-scaled

# Attribute Types

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- **Nominal (categorical):** means “relating to names.”
- The values of a **nominal attribute** are symbols, *names of things*, categories and states.
- *Hair\_color* = {auburn, black, blond, brown, grey, red, white}
- Marital status, occupation, ID numbers, zip codes etc
- **Binary**
  - Nominal attribute with only 2 states (0 and 1)
- Where 0 typically means that the attribute is absent, and 1 means that it is present.
- Also referred to as **Boolean** if the two states correspond to *true* and *false*.

# Attribute Types

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- Symmetric binary: Both outcomes equally important, carry the same weight. e.g., gender
- Asymmetric binary: outcomes not equally important.
  - e.g., medical test (positive vs. negative)
  - Convention: assign 1 to most important outcome (e.g., HIV positive)
- **Ordinal**
  - Values have a meaningful order (ranking) but magnitude between successive values is not known.
  - *Size* = {*small, medium, large*}, grades, army rankings

# Numeric Attribute Types

- **Numeric Attributes (integer or real-valued)**
  - Interval-scaled
  - Ratio-scaled
- **Interval**
  - Measured on a scale of **equal-sized units**
  - Values have order
    - E.g., *temperature in C° or F°, calendar dates*
    - *0°C does NOT mean “no temperature”*  
*✗ 20°C is NOT twice as hot as 10°C ✗*



# Numeric Attribute Types

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## ■ Ratio

- Inherent **zero-point** (Zero represents the **complete absence** of the quantity being measured)
- We can speak of values as being an order of magnitude larger than the unit of measurement (10 K° is twice as high as 5 K°).
- Weight 0 kg means no weight
- 10 kg is twice 5 kg

# Discrete vs. Continuous Attributes

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- **Discrete Attribute**

- Has only a finite set of values
  - E.g., zip codes, profession, or the set of words in a collection of documents
  - {44000, 54000, 60000} Even though they are numbers, they are:
  - Labels
  - Not continuous quantities
  - Finite possible value
- Sometimes, represented as integer variables
- Note: Binary attributes are a special case of discrete attributes: {0,1 } Only two possible values → finite → countable → discrete

# Discrete vs. Continuous Attributes

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- **Continuous Attribute**

- Has real numbers as attribute values
  - E.g., temperature, height, or weight
  - 25°C, 25.1°C, 25.12°C, 25.123°C ... Between 25 and 26, there are infinitely many possible values.
- Practically, real values can only be measured and represented using a finite number of digits
- Continuous attributes are typically represented as floating-point variables